

DESIGN & SETUP REPORT

A transcranial focused-ultrasound targeting pipeline for the squirrel monkey: a *Saimiri sciureus* museum microCT registered to the VALiDATE29 brain atlas, with a hard CT-calibration guard (deliverables 1–2)

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Abstract. We document the fifth pillar of TUBA — a transcranial focused ultrasound (FUS) pipeline for the common squirrel monkey (*Saimiri sciureus*), prepared for NIH R01 EB037345 (Aim 3): per-animal skull models for time-reversal aberration correction and multifocal beam design targeting the hand representations of primary somatosensory (S1 areas 3b and 1) and primary motor (M1, area 4) cortex. This report covers **deliverables 1–2 (the geometry stages)** and is scoped, by design decision, to a *geometry-only* build: no HU-calibrated colony CT is available yet, so the pillar is bootstrapped on an *uncalibrated* museum microCT (squirrel monkey *Saimiri sp.*, NMNH USNM 194346, MorphoSource media 000116521, UTCT; 490 slices, 0.0977 mm in-plane $\times$ 0.1189 mm slice) registered to the VALiDATE29 multi-channel MRI atlas of the squirrel-monkey brain (Schilling *et al.* 2017; 29 animals, histologically-defined cortical labels). **Both the skull and the atlas are staged and the geometry is now measured end-to-end:** the anisotropic microCT is isotropised to 0.0977 mm, its orientation is confirmed clean RAS, a *solid* endocranial cavity is extracted (15.9 mL), and the `cavity_binary` SyN to the atlas brain mask fits to Dice 0.96, warping the region-level labels (confirmed *region-level* — S1 = anterior parietal cortex, M1 = primary motor cortex, l/r separate) back to subject-space S1/M1 targets that are anatomically correct (M1 rostral and dorsal to S1, 4.5 mm apart). The pillar reuses TUBA's core primitives from the mouse/rat/macaque pillars almost unchanged — bone-shell cavity extraction (`tuba.core.cavity.bone_shell_cavity`), `cavity_binary` ANTs SyN (affine + deformable), atlas-into-subject warp, and the beam-aligned (c, ρ) slab export. Its *distinctive* contribution is the **CT-calibration guard** for deliverable 1: a new module `tuba.core.hu_acoustics` implements the Aubry (2003) porosity $HU \rightarrow (\rho, c, \alpha)$ model but *refuses to run* on input not flagged Hounsfield-calibrated (`UncalibratedInputError`), so a museum microCT can never silently produce quantitatively wrong acoustics; an explicitly-flagged placeholder ramp is the sanctioned geometry-only fallback. Deliverable 1 also reports the points-per-wavelength (PPW) budget across the 2–4 MHz operating band on the simulation grid, and a grid-convergence run of the transcranial time-of-flight aberration (the scalar a time-reversal correction must undo), which converges to 0.24% between 60 and 30 μ m. The scan-dependent constants (orientation flips, intensity thresholds, cavity hull geometry) are now *calibrated* against the staged scan, and the cavity, registration, target, and placement outputs and their QC figures are produced by the end-to-end driver. The downstream wave physics (deliverables 3–7: time-reversal, multifocal synthesis, the two-foci feasibility sweep, MI/bioheat safety, and the per-animal export artifact) is out of scope for this branch and consumes the (c, ρ, dz) slab in the `fullwave2-*` solver siblings. Code: `src/tuba/species/saimiri.py`, `tuba.core.hu_acoustics`, `tuba.atlases.validate29`; data manifest + fetchers at `src/tuba/data/`.

1 Scope and status

Like the four prior TUBA pillars (mouse, macaque, human, rat), this one is now calibrated end-to-end against a staged scan — with a measured cavity volume, a fitted registration, and subject-space targets — but with one deliberate boundary: the acoustics are *geometry-only*. The funded project targets a Vanderbilt colony whose HU-calibrated CT is not yet in hand, so we build and verify the full geometry on a public museum microCT and gate the quantitative acoustics behind a calibration guard (§4) so nothing downstream mistakes an uncalibrated skull for a quantitative one. Two decisions frame the branch:

1. **Subject data:** a public museum microCT (§2), geometry only. The $HU \rightarrow$ acoustics step refuses to run on it.
2. **Scope:** deliverables 1–2 (CT $\rightarrow$ acoustic model + atlas registration + target export + QC). Deliverables 3–7 (the wave-solver study) are out of scope here and live in the `fullwave2-*` siblings (§7).

Both the museum skull (§2) and the VALiDATE29 atlas (§3) are staged; all geometry numbers below are measured from them.

2 Source data

Skull. The skull volume is a microCT scan of a squirrel monkey (*Saimiri sp.*) cranium, museum specimen USNM 194346 (National Museum of Natural History), scanned at the University of Texas High-Resolution X-ray CT Facility (UTCT, ACTIS scanner) and distributed as MorphoSource media 000116521 [1] (also on

DigiMorph [2]). The reconstruction is 490 axial slices, 504×422 px, 16-bit, *anisotropic* at 0.0977 mm in-plane $\times$ 0.1189 mm slice — a dry cranium, no soft tissue. Part of the openVertebrate / oUTCT collection the TUBA macaque pillar [10] also draws from (Copes *et al.* 2016 [3]). The MorphoSource record’s taxonomy is internally inconsistent (catalogue field *S. boliviensis*; Rossie’s scan description *S. sciureus sciureus*, male) — genus *Saimiri* either way. Table 1 summarises the parameters.

Table 1. Saimiri USNM 194346 (MorphoSource media 000116521) scan parameters, from the media manifest. The intensity axis is *uncalibrated* (no HU phantom in the field of view).

Specimen	<i>Saimiri sp.</i> , USNM 194346 (NMNH)
MorphoSource media	000116521 (ark:/87602/m4/M116521)
Facility	UT Austin High-Resolution X-ray CT (UTCT ACTIS)
Slices	490 axial, 504×422 px, 16-bit
Voxel	0.0977 mm in-plane $\times$ 0.1189 mm slice (anisotropic)
Working pitch	0.0977 mm isotropic (slice axis up-sampled)
Calvarium thickness	1.0–1.5 mm (~ 10 – 15 voxels)
Intensity	uncalibrated 16-bit reconstruction counts

Anisotropy $\rightarrow$ isotropic working cache. The shared cavity/align/slab tools (calibrated on the isotropic rat/mouse museum scans) assume isotropic voxels, so `downsample_to_working` isotropises the staged stack: it reformats to RAS at the in-plane pitch with the shared TIFF streamer, then up-samples *only* the coarse slice axis (0.1189 $\rightarrow$ 0.0977 mm, order-1) to a 0.0977 mm isotropic cache. Up-sampling (never down-sampling) that axis cannot erode the thin cortical-bone shell, and the whole downstream isotropic pipeline is then exact, with no change to the shared code.

Uncalibrated intensity — the central caveat. Museum microCT carries no hydroxyapatite/HU phantom, so voxel values are arbitrary reconstruction counts, not Hounsfield units. Every prior TUBA pillar faces the same issue and handles it by anchoring a histogram-relative intensity $\rightarrow$ (c, ρ) ramp on the cortical-bone mode [9, 10]: robust to detector scale, but explicitly *not* quantitatively calibrated. For the squirrel monkey we make that limitation a hard boundary rather than a footnote (§4): the scan is flagged `UNCALIBRATED_MICROCT` and quantitative acoustics are refused.

Wavelength / PPW criterion. The neuromodulation operating point is $f_0 = 2$ MHz with a 2–4 MHz design range. The most restrictive wavelength is in the water column at 4 MHz: $\lambda_{\min} = c_{\text{water}}/f = 1500/4 \times 10^6 = 0.375$ mm. A finite-difference solve wants $\gtrsim 6$ points per wavelength there, so the simulation grid pitch is set to $\Delta x = 60 \mu\text{m}$ (`SIM_DX_M` in `tuba.species.saimiri`); the resulting PPW budget is Table 2. Note the 0.0977 mm microCT working cache is *coarser* than the $60 \mu\text{m}$ sim grid, so the skull is up-sampled onto the sim grid by the slab sampler (§6) — the geometry is faithful; only the absolute acoustics are placeholder.

Table 2. Points-per-wavelength budget on the $\Delta x = 60 \mu\text{m}$ simulation grid, using the shortest (water) wavelength at each frequency (`tuba.core.hu_acoustics.report_ppw`). All three bands clear the 6-PPW floor.

Frequency	λ_{water} (mm)	PPW @ $60 \mu\text{m}$
2.0 MHz	0.750	12.50
3.0 MHz	0.500	8.33
4.0 MHz	0.375	6.25

3 VALiDATE29 reference atlas

The brain anatomy is provided by the VALiDATE29 atlas of the squirrel monkey brain [4], built from 29 animals at the Vanderbilt University Institute of Imaging Science and distributed CC BY via NITRC (RRID:SCR_015542). Unlike the subject skull, **the atlas is staged and validated in this report**: `tuba.data.fetch_validate29` discovers the current file-release link from the NITRC project page (reusing the WHS click-through of the rat pillar [11]) and unpacks it under `$VALIDATE29_DEST`; the fetch was exercised end-to-end here (v1.2 release, 30.4 MB). Every volume is 212^3 at 0.300 mm isotropic. VALiDATE29 is multi-channel: in-vivo anatomical templates (T_2 , T_1 , proton-density), diffusion templates (FA, MD), an ex-vivo structural + FA template, a *shipped brain mask* (`VALiDATE-brainmask`, ~ 33 mL) and a discrete cortical + white-matter label volume (`VALiDATE3-labels`; 86 label ids) co-registered from Nissl/myelin sections. It supersedes the earlier single-subject squirrel-monkey MRI atlas [5]. Figure 1 shows the T_2 template, brain mask, and cortical labels through the brain-mask centroid.

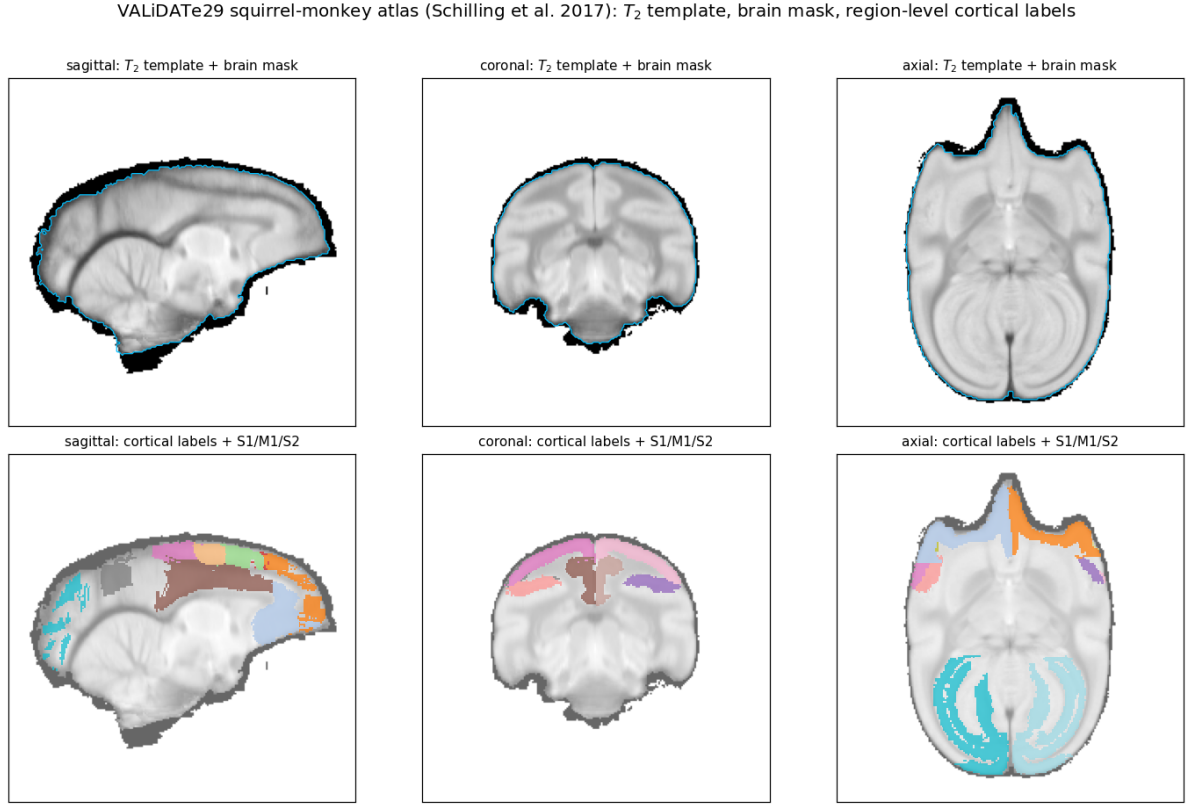


Figure 1. VALiDATE29 squirrel-monkey atlas through the brain-mask centroid (staged v1.2). **Top:** T_2 template with the shipped brain-mask contour (blue). **Bottom:** region-level cortical labels (gray-matter ids 1–18) over the template, with the S1 (anterior parietal cortex), M1 (primary motor cortex), and S2 (PV/S2) target markers where the centroid slice intersects them. The squirrel monkey is near-lissencephalic, so the cortical areas form broad dorsolateral bands. Generated by `docs/saimiri/make_figures.py` from the staged atlas.

Layout tolerance (a real calibration note). The staged VALiDATE29.zip double-nests VALiDATE29/VALiDATE29/ and carries macOS `__MACOSX/.` resource-fork cruft, and its lookup table is a plain space-delimited id name ABBREV text file (not ITK-SNAP format) sitting beside a `LICENSE.txt`. The binding `tuba.atlases.validate29.VALiDATE29` walks the tree recursively, *ignores the cruft*, resolves the template/label/mask by keyword, prefers a label-named table over `LICENSE`, and raises with the candidate list rather than guessing — all confirmed against the staged files. The cavity-binary registration uses the shipped brain mask directly as its fixed image (no derive step, no ANTs). Paths are overridable by constructor argument or environment variable (`VALIDATE29_DEST`, `VALIDATE29_URL`).

Target structures (confirmed against the staged LUT). The study targets the *hand* representations in S1 (areas 3b, 1) and M1; S1 and M1 are only $\sim 2\text{--}3\text{ mm}$ apart in the squirrel monkey, which is what makes the downstream two-foci feasibility question (§7) non-trivial. Inspecting the staged label table settles an important granularity point: **VALiDATE29’s histological parcellation is region-level, not Brodmann-area-level.** There is no separate “area 3b” / “area 1” / “area 4” label; the primary somatosensory hand areas 3a/3b/1/2 are bundled into a single *anterior parietal cortex* (APC) label and the primary motor cortex is one label, with left and right as *separate* label ids. The binding maps the study targets accordingly (S1 $\rightarrow$ APC, M1 $\rightarrow$ primary motor, S2 $\rightarrow$ PV) and is hemisphere-aware. The atlas-space centroids measured from the staged label volume (Table 3) are anatomically consistent: M1 sits rostral and slightly dorsal to S1 across the central sulcus, with clean L/R mirror symmetry. Splitting 3b from area 1, or isolating the hand knob within APC, needs a stereotaxic prior on the homunculus that the distributed label volume does not carry (§9).

Table 3. VALiDATE29 study-target labels and their *atlas-space* centroids (left hemisphere; right is the mirror), measured from the staged VALiDATE3-labels volume via `tuba.atlases.validate29`. Coordinates in the atlas RAS frame (mm).

Target	Atlas label (l/r id)	centroid (x, y, z)	vox	mL
S1	anterior_parietal_cortex (11/12)	(−10.1, 1.1, 10.3)	18 166	0.49
M1	primary_motor_cortex (3/4)	(−7.4, 5.8, 12.0)	11 810	0.32
S2	parietal_ventral/ S2 (7/8)	(−13.5, 3.7, 4.6)	8 755	0.24

4 Deliverable 1: CT → acoustic model, with a calibration guard

This is the pillar’s distinctive contribution and the reason a geometry-only build is safe. The module `tuba.core.hu_acoustics` provides a *quantitative* HU mapping and a hard guard around it.

Porosity model. For Hounsfield-calibrated input we use the Aubry (2003) [6] porosity model. Bone porosity is estimated from CT radiodensity,

$$\Phi(\text{HU}) = 1 - \text{clip}\left(\frac{\text{HU} - \text{HU}_{\text{water}}}{\text{HU}_{\text{bone}} - \text{HU}_{\text{water}}}, 0, 1\right), \quad (1)$$

with $\Phi = 1$ water/marrow and $\Phi = 0$ dense cortical bone, and the acoustic properties interpolate linearly in the bone fraction $(1 - \Phi)$:

$$\rho(\Phi) = \rho_{\text{water}} + (\rho_{\text{bone}} - \rho_{\text{water}})(1 - \Phi), \quad (2)$$

$$c(\Phi) = c_{\text{water}} + (c_{\text{bone}} - c_{\text{water}})(1 - \Phi), \quad (3)$$

with cortical endpoints $\rho_{\text{bone}} = 1900 \text{ kg/m}^3$, $c_{\text{bone}} = 2900 \text{ m/s}$ consistent with the rat/macaque pillars, and water $\rho = 1000 \text{ kg/m}^3$, $c = 1500 \text{ m/s}$. Attenuation is taken proportional to bone fraction between a soft-tissue floor (0.5 dB/cm/MHz) and a cortical-bone value (8 dB/cm/MHz) [7, 8], reported in dB/cm/MHz and applied by the downstream solver. SI units throughout.

The guard. `AubryHUMapping.properties()` raises `UncalibratedInputError` unless the bound `CTCalibration.kind == 'hu'`. The squirrel-monkey scan is a museum microCT with no phantom, so the pillar binds `CT_CALIBRATION = UNCALIBRATED_MICROCT` and the quantitative path is refused by construction. Rather than silently emitting physically meaningless densities and sound speeds from arbitrary reconstruction counts, the caller must either (a) supply an HU-calibrated colony CT — one line, `CT_CALIBRATION = CTHCalibration(kind='hu', hu_bone_max=(scan))` — which unlocks the quantitative mapping automatically, or (b) opt in explicitly to `placeholder_ramp()`, an `intensity → (c, ρ)` ramp carrying nominal cortical endpoints and a `.placeholder = True` tag so downstream code can detect the uncalibrated origin. The slab loader (§6) uses the flagged placeholder ramp: geometry is faithful, absolute (c, ρ) are nominal.

PPW report + grid convergence. `acoustic_model_report()` prints the calibration state, the PPW table (Table 2) at 2/3/4 MHz on the sim grid, and — given a placement and a staged scan — one grid-convergence run of the transcranial *time-of-flight aberration*

$$\tau = \int \left(\frac{1}{c(x)} - \frac{1}{c_{\text{ref}}} \right) dx, \quad (4)$$

the phase-insertion delay that a time-reversal / phase-conjugation correction must undo, computed along the central beam ray at Δx and $\Delta x/2$ and reported as an absolute and relative change. On the staged skull, along the beam to the M1 target, this gives $\tau = -630.2 \text{ ns}$ at $60 \mu\text{m}$ and -631.7 ns at $30 \mu\text{m}$ — a 1.5 ns (0.24 %) change, i.e. the property model is converged at the sim pitch. This is a convergence of the acoustic-*property* model with voxel pitch, not of a full-wave solve (which lives in the solver siblings, §7); it is exactly the quantity the downstream aberration correction consumes, so it is the right model-side convergence to report here. The units and sign are checked in the offline test suite (homogeneous medium $\Rightarrow \tau = 0$; a faster bone slab $\Rightarrow \tau < 0$; a smooth profile converges as $\Delta x \rightarrow 0$).

5 Deliverable 2: geometry — cavity, registration, targets

The geometry stages reuse the rat/macaque dry-skull spine with no algorithmic change; only the species constants differ, and they are now calibrated against the staged scan. Figure 2 shows the result.

Saimiri USNM 194346 microCT + solid endocranial cavity (15.9 mL) — 97.7 μ m isotropic, uncalibrated (geometry only)

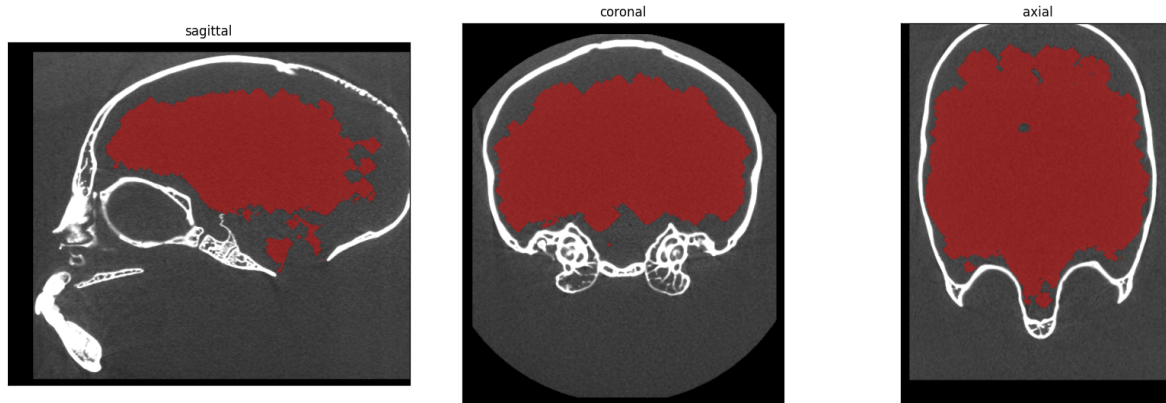


Figure 2. Saimiri USNM 194346 microCT (97.7 μ m isotropic working cache) with the extracted *solid* endocranial cavity (red, 15.9 mL) through the cavity centroid. Sagittal, coronal, and axial. The cavity fills the braincase from the frontal to the occipital pole and the cerebellar fossa, follows the inner table, is bilaterally symmetric, and excludes the orbits — the shape the registration needs. Uncalibrated intensity (geometry only). Generated by `docs/saimiri/make_figures.py` from the build cache.

Ingest + orientation (confirmed). The MorphoSource TIFF stack carries no DICOM affine; as in the rat pillar [11] the storage→anatomy mapping is fixed by an orientation probe (mid-stack orthoslices). The rat/macaque museum-CT defaults in `tuba.species.saimiri` (`SWAP_ROW_COL=True`, `VOXEL_SIGNS=(-1, -1, +1)`, storage (slice=*AP*, row=*DV*, col=*LR*)) transfer correctly to this UTCT stack: the probe confirms a clean RAS frame — anterior (snout, orbits) at $+y$, dorsal (braincase vault) at $+z$, bilaterally symmetric about x , right-handed — with no residual tilt (align angles 0°, as in rat). True L/R handedness is not recoverable from a bare museum skull (no fiducial); $+x$ is taken as Right per RAS and the hemisphere labels carry that caveat.

Endocranial cavity (measured, solid). The many dry-cranial openings (orbits, optic canals, foramen magnum) rule out the mouse-style pure-hysteresis extractor, exactly as in rat and macaque, so we use the species-agnostic bone-shell extractor `tuba.core.cavity.bone_shell_cavity` (per-column basicranium floor + per-row plugs + per-axial hull), with a squirrel-scale hull envelope (`HULL_X_HALF_MM=24`, tighter rostral cutoff at the basicranium than the vault). One calibration point was specific to this scan and mirrors how the rat’s naturally-watertight 8-bit base behaves: at the histogram bone knee (~ 6000 counts) the thin *midline* basicranium drops below threshold, the per-column floor fails to seal it, and “outside” floods up the midline into the braincase centre (a hollow-centred cavity). Dropping the cavity shell threshold to a watertight 5000 counts (`CAVITY_BONE_LOW`, below the bone knee kept for the ramp/centroid) seals the floor and yields a *solid* cavity (central-fill 0.97 vs 0.26), verified by a fill-holes check. The extracted volume is 15.9 mL — a conservative, inner-table-inclusive figure within the ~ 16 –26 mL comparative range for *Saimiri*; because the cavity is only the registration moving image, the affine to the 33 mL atlas brain mask absorbs the individual scale difference.

Registration — cavity_binary SyN (affine + deformable). Deliverable 2 calls for affine *and* deformable registration, so — unlike the macaque/rat pillars, which stop at a 12-DOF Affine — we use the full `cavity_binary` SyN path (`tuba.core.warp.register_subject_to_atlas`, fixed = VALiDATE29 brain mask, moving = subject cavity), the same convention the mouse and mini pillars use. The atlas template and cortical labels are then pulled back into subject space through the saved inverse transforms (`warp_atlas_into_subject`, linear for the template, nearest-neighbour for the labels). Unlike the rat and mouse pillars, which *derive* a brain mask from the annotation, VALiDATE29 ships its own (VALiDATE-brainmask), so the binding uses it directly as the SyN fixed image — no derive step and no ANTs for the mask. On the staged scan the SyN converges in ~ 1 min; the fit is excellent — the atlas brain mask, warped back into subject space, overlaps the extracted cavity at **Dice 0.962** (15.9 vs 16.2 mL) and fills the braincase following the inner table (Fig. 3).

Target export + QC (deliverable 2 outputs). `export_targets()` writes the S1 (anterior parietal cortex) and M1 (primary motor cortex) coordinates in the subject simulation frame to `saimiri_targets.json`,

alongside the transform prefix (saved under `$TUBA_SAIMIRI_REG_DIR` as `saimiri_cavity_to_validate29`), and `qc_figure()` writes the three-panel overlay of Figure 3. The subject-space left-hemisphere centroids (Table 4) are anatomically correct: **M1 sits 3.7 mm rostral and 1.2 mm dorsal to S1**, 4.5 mm centroid-to-centroid, both in the left hemisphere — the sensorimotor topology the study depends on, carried through the cavity registration from the confirmed atlas-space labels (Table 3).

Table 4. Study-target centroids in the *subject* aligned-native RAS frame (left hemisphere), from `saimiri_targets.json` — the atlas labels of Table 3 warped through the cavity SyN. Volumes scale with the cavity/atlas affine.

Target	Atlas label (id)	subject centroid (x, y, z)	mL
S1	anterior_parietal_cortex (11)	(−8.6, −4.1, 14.3)	0.23
M1	primary_motor_cortex (3)	(−6.2, −0.5, 15.5)	0.15



Figure 3. Deliverable-2 QC: VALiDATE29 atlas registered into the Saimiri USNM 194346 skull. Bone shell (gray), the warped VALiDATE29 brain mask (cyan) filling the braincase, and the S1/M1 target markers, in sagittal, coronal, and axial. M1 lies rostral and dorsal to S1 in the left hemisphere across all three planes. Cavity vs warped-brain Dice = 0.96. Generated by `docs/saimiri/make_figures.py / tuba.species.saimiri.qc_figure`.

6 Placement and slab export

`place_on_skull` computes the apex/target/beam triple for a focused bowl aimed at a study target, and `load_slab` returns ($c, \rho, dz_{\text{slab}}, \text{frame}$) with the *same signature* as the mouse/rat/macaque loaders, so any FDTD or angular-spectrum driver wired against one species runs on this one unchanged. On the staged skull, aiming at the M1 target, it lands the bowl apex on the outer-bone surface, lifts it 11.2 mm for scalp clearance, and returns a 20 mm apex-to-target standoff with the beam axis $\hat{n} = (-0.12, 0.09, -0.99)$ (near-vertical, entering the dorsal vault). The slab ramp is the flagged placeholder (§4); the cavity is overwritten with brain-parenchymal properties ($c = 1540$ m/s, $\rho = 1040$ kg/m³) exactly as in the other pillars. This slab is the hand-off boundary to the wave-solver siblings.

7 Out of scope here: the wave-physics study (deliverables 3–7)

TUBA is the geometry/atlas/targeting layer; the wave physics lives in the `fullwave2-*` FDTD siblings and consumes the (c, ρ, dz) slab and placement this pillar exports. The funded study’s remaining deliverables are therefore deliberately *not* in this branch:

- D3 time-reversal aberration correction through the individual skull (virtual source at each target → recorded element signals → phase-conjugate transmit delays);
- D4 multifocal beam synthesis — two simultaneous foci (S1 + M1) with independent amplitude control, plus at least one alternative to superposed time reversal (e.g. Gerchberg–Saxton);
- D5 the feasibility study (highest-priority result): sweep two-foci separation 1.5–4.0 mm at 2, 3, 4 MHz through the skull model, reporting −6 dB focal dimensions, peak-to-valley contrast, off-target ratio, and transcranial focal gain;
- D6 safety margins — MI at target and skull, and a bioheat estimate over the 400 kPa / 300 ms / 3 s duty cycle;
- D7 the per-animal targeting artifact (element delays, apodisations, target metadata, safety numbers).

The PPW budget (Table 2) and the time-of-flight grid-convergence (§4) are the model-side inputs D3 and D5 will build on.

8 Verification

Verification is at two levels: (i) the end-to-end build now runs on the staged skull + atlas (`python -m tuba.species.saimiri build`), producing the cavity, registration (Dice 0.96), subject-space targets, QC figure, placement, and D1 report of this document; and (ii) offline unit tests (`tests/test_saimiri.py`, 11 tests, no network / no antspyx / no data) that pin the logic which must hold regardless of a scan:

- importing the pillar stays ANTs-free (ANTs is lazy inside `tuba.core.warp` and the atlas binding);
- the calibration guard refuses uncalibrated input on every entry point, and the calibrated Aubry mapping is physically monotonic (porosity $\downarrow$, $\rho/c/\alpha \uparrow$ with HU) with the correct water / cortical endpoints and clamping;
- the placeholder ramp is flagged and maps to nominal endpoints;
- the PPW, time-of-flight-aberration, and grid-convergence maths (homogeneous $\Rightarrow 0$; bone slab $\Rightarrow$ analytic; smooth profile converges);
- VALiDATE29 file/label resolution and S1/M1 target keyword mapping on a synthetic atlas, plus a synthetic affine registration round-trip;
- the data manifest carries the three Saimiri entries with the expected calibration/license semantics.

The full repository suite is 41 passing + 1 skipped; `ruff` is clean.

9 Open items

The geometry is now staged and validated end-to-end (skull: §5, Figs. 2–3; atlas: §3, Fig. 1). The remaining items are the acoustic calibration and the downstream study:

1. **Quantitative acoustics.** When a Vanderbilt colony CT is available, set `CT_CALIBRATION` to an hu calibration to unlock the quantitative Aubry mapping (one line), and re-run the grid-convergence with the real property field. Until then the acoustics stay guarded and the slab ramp is the flagged placeholder.
2. **Hand sub-representation.** Add a stereotaxic hand-knob prior within anterior parietal cortex (the label granularity limit of §3) to resolve areas 3b/1 from the bundled APC label; the region→label mapping itself is confirmed and pinned.
3. **L/R handedness.** A museum skull carries no L/R fiducial; confirm the hemisphere convention against a colony CT with known laterality before per-animal targeting.
4. **Downstream study.** Hand the exported slab + targets to the `fullwave2-*` siblings for deliverables 3–7 (§7).

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